REPORT

Submitted by:-
D.Kusuma Sree

Project Title:-
Real time Google Play store data analytics - Python

Internship Type:-
Online

Tools Used:-
Python,pandas,plotly,google colab,git hub.

Internship last date:-
17th march 2026

Submitted to:-
Elevance Skills

Submission Date:-
12-03-2026

Introduction:-

In this project I used python and plotly to create different charts to understand the google play store dataset.The Google Play Store contains thousands of applications from different categories such as games, education, entertainment, and business. Analyzing this data can help us understand user preferences, app popularity, and performance trends.The main aim of this project is to analyze the Google Play Store dataset and create different types of data visualizations using Python and Plotly. Various charts such as bar charts, bubble charts, line charts, choropleth maps, and stacked area charts are used to represent the data in a clear and interactive way.

By applying different filters and conditions, the visualizations help in understanding patterns in app installs, ratings, reviews, and categories. This project helps in learning how data visualization can be used to transform raw data into meaningful insights.

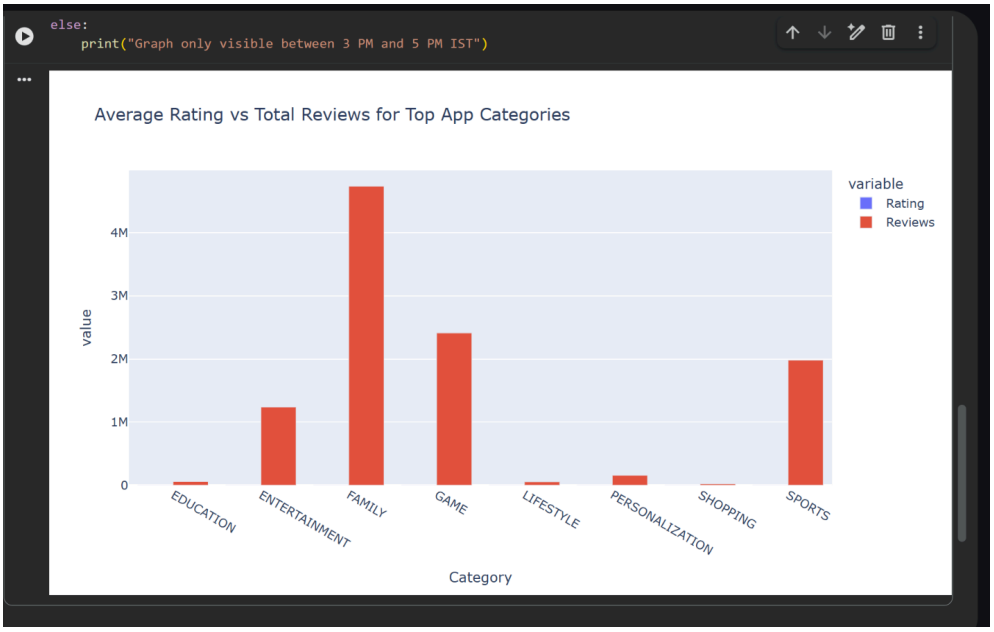
Task 1: Grouped Bar Chart – Average Rating vs Total Reviews

Objective

To compare the average rating and total number of reviews for the top 10 application categories using a grouped bar chart.

Filters Applied

- App size greater than 10 MB
- Last updated in January
- Average rating greater than or equal to 4.0



Analysis

The grouped bar chart allows comparison between rating performance and review volume across categories. Categories with both high ratings and large review counts indicate strong user satisfaction and engagement.

Task 2: Choropleth Map – Global Installs by Category

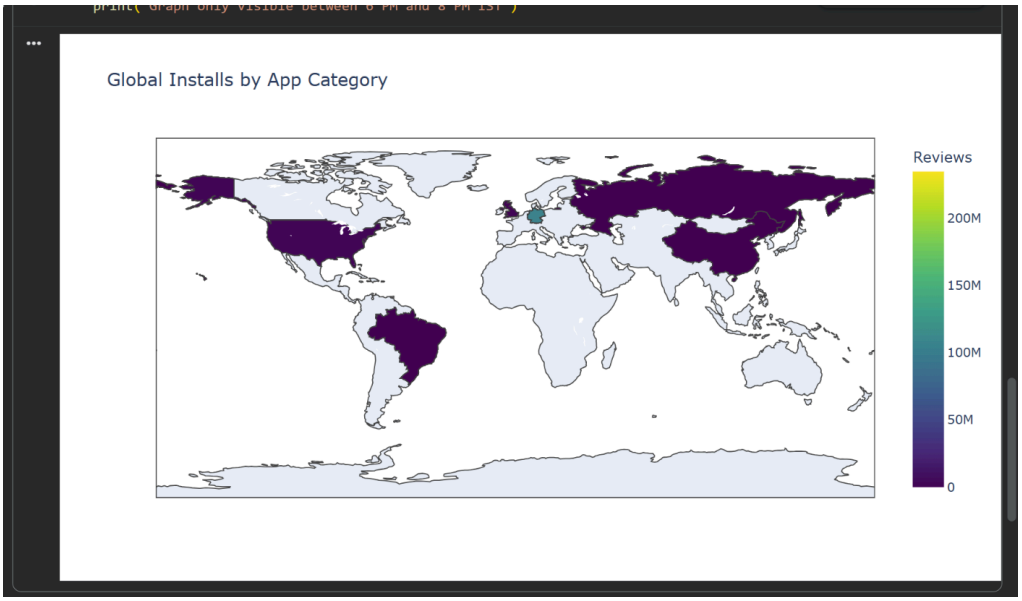
Objective

To visualize the geographic distribution of application installs across different categories using an interactive choropleth map.

Filters Applied

- Only the **top five categories** by installs
- Categories not starting with **A, C, G, or S**
- Categories with installs greater than **1 million** highlighted

Output Visualization



Analysis

The choropleth map highlights regional patterns in app installations, enabling identification of categories that perform well globally.

Task 3: Dual Axis Chart – Installs vs Revenue

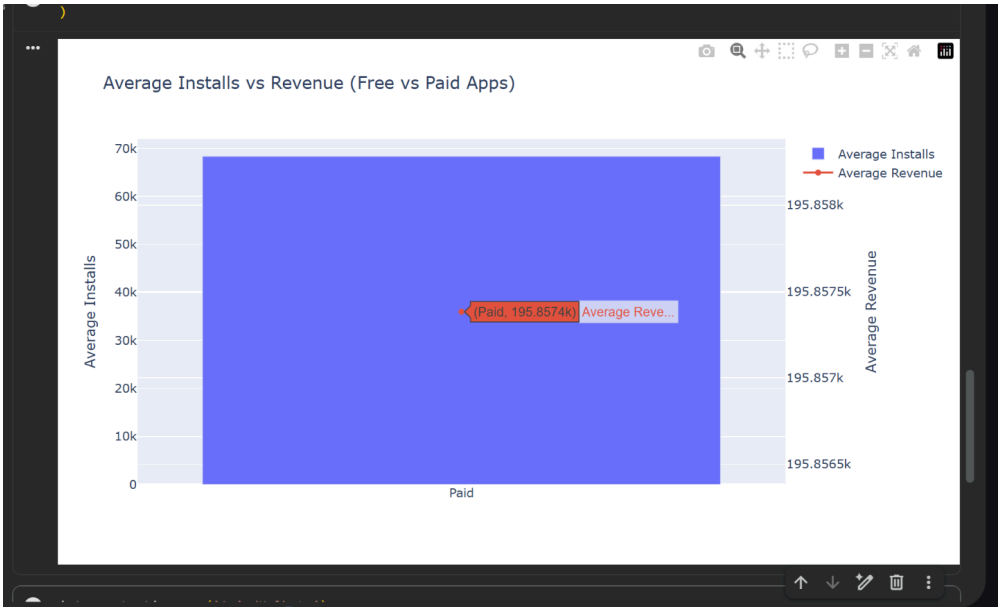
Objective

To analyze the relationship between **average installs and revenue** for free and paid applications.

Filters Applied

- Installs greater than **10,000**
- Revenue greater than **$10,000**
- Android version greater than **4.0**
- Application size greater than **15 MB**
- Content rating set to **Everyone**
- Application name length less than **30 characters**

Output:-



Analysis

The dual-axis visualization effectively compares install counts and revenue generation for different app types, highlighting the performance differences between free and paid applications.

Task 4: Time Series Line Chart – Install Trend Analysis

Objective

To analyze how application installs evolve over time for selected app categories.

Filters Applied

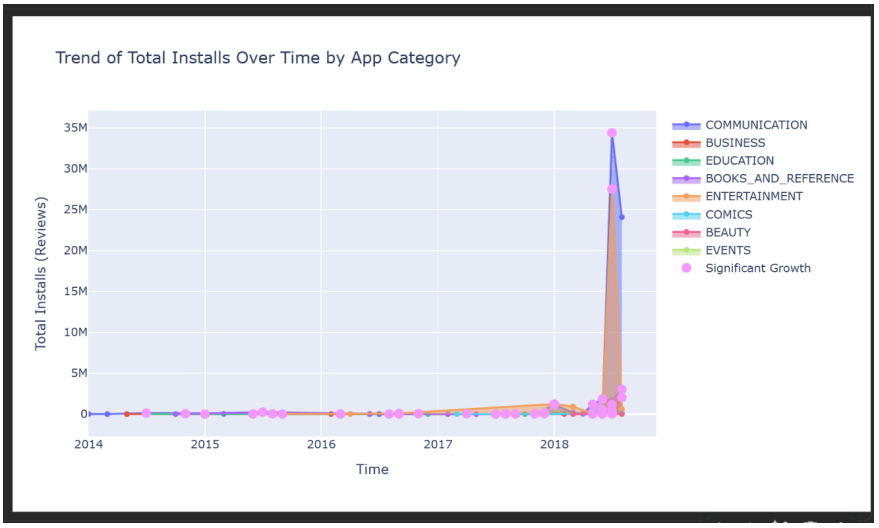
- Reviews greater than **500**
- Categories beginning with **E, C, or B**
- Application names not starting with **X, Y, or Z**
- Application names not containing the letter **S**

Additional Features

Category translations applied for visualization:

- Beauty → **Hindi**
- Business → **Tamil**
- Dating → **German**

Output:-



Analysis

The time series chart highlights installation trends and identifies periods of significant growth across categories.

Task 5: Bubble Chart – App Size vs Rating

Objective

To explore the relationship between **application size and average rating**, with bubble size representing installs.

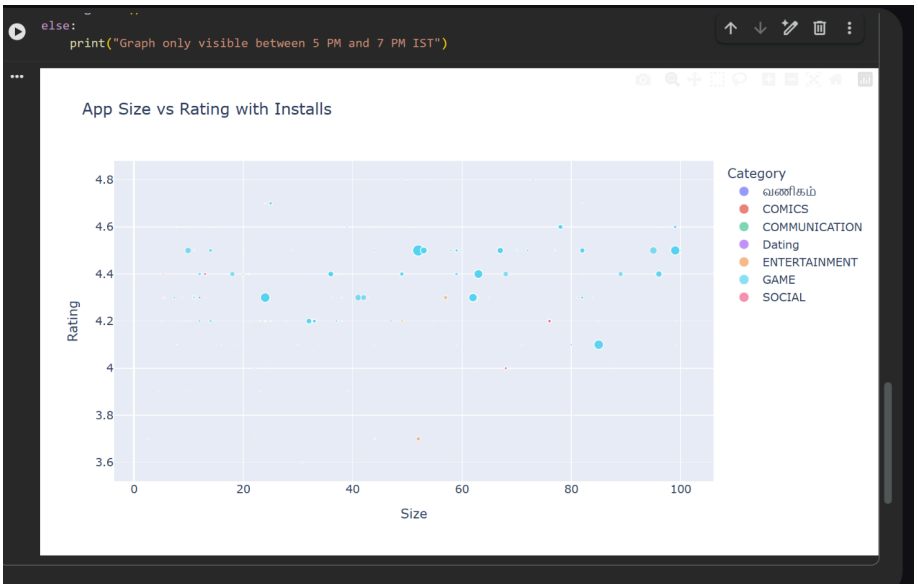
Filters Applied

- Rating greater than **3.5**
- Reviews greater than **500**
- Installs greater than **50,000**
- Selected categories including Game, Beauty, Business, Comics, Communication, Dating, Entertainment, Social, and Events

Additional Features

- Game category highlighted in **pink**
- Category translations applied

Output:-



Analysis

The bubble chart provides a visual representation of how application size and rating influence popularity and user engagement.

Task 6: Stacked Area Chart – Cumulative Install Trends

Objective

To visualize cumulative installs over time across different app categories using a stacked area chart.

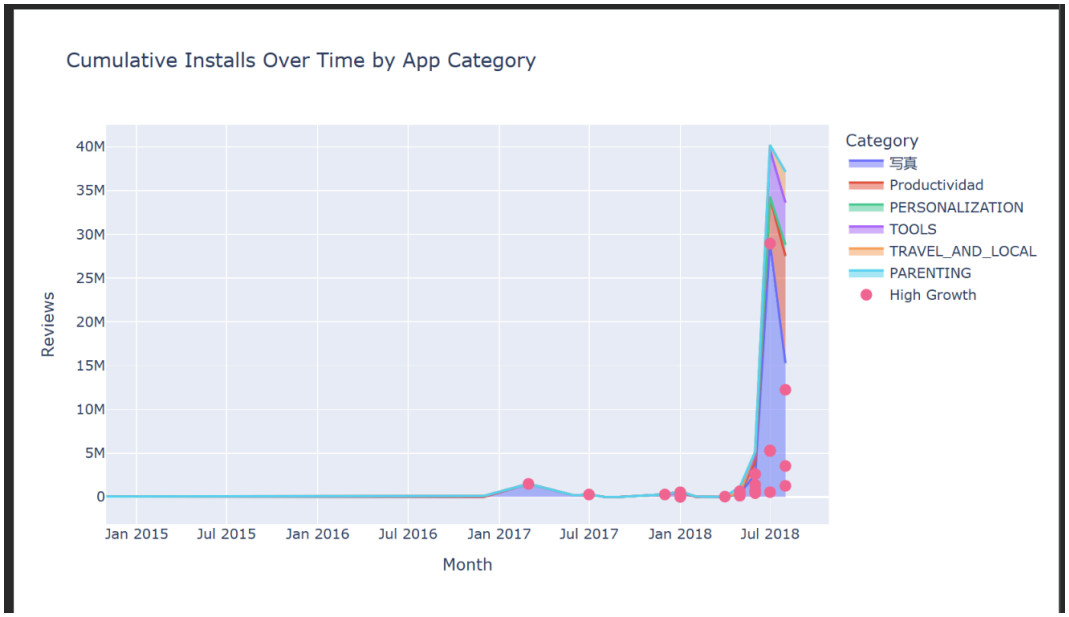
Filters Applied

- Average rating greater than or equal to **4.2**
- Reviews greater than **1,000**
- Application names without numeric characters
- Categories beginning with **T or P**
- Application size between **20 MB and 80 MB**

Legend Translations

- Travel & Local → **French**
- Productivity → **Spanish**
- Photography → **Japanese**

Output:-



Analysis:-

The stacked area chart demonstrates cumulative installation trends, allowing easy comparison of category growth patterns over time

Conclusion:-

The project successfully demonstrates the use of interactive data visualization techniques to analyze application data from the Google Play Store dataset. By applying multiple filters, language translations, and time-based display conditions, the visualizations provide deeper insights into app performance, category trends, and user engagement. These techniques can be extended to build real-time analytical dashboards for mobile application marketplaces.